



ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your Brain *Health*, Ultimate Guide

Clearer Thinking, a Steadier Mood, and the Treatable Causes Worth Ruling Out First

A guide for adults noticing brain fog, memory that feels less sharp than it used to be, or persistent anxiety and low mood, who want to understand the terrain underneath those symptoms, alongside whatever medical or mental health care they are already receiving.

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BEFORE YOU START

A note before you start. This guide is educational information about the biology behind brain fog and mood symptoms, and the general categories of evaluation and support used to address treatable contributors to both. It is not individualized medical, psychiatric, or psychological advice, it does not diagnose any condition, and it does not replace your relationship with your physician, psychiatrist, therapist, or my team. If you are currently in therapy or under psychiatric care, please continue exactly as prescribed and treat everything here as additive, not a replacement. Before anything else, the section below covers what to do in a crisis.

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If You Are in Crisis Right Now

If you are having thoughts of harming yourself or of suicide, please stop reading and reach out immediately. Call or text **988** to reach the Suicide and Crisis Lifeline in the United States, available 24 hours a day, every day. If you or someone else is in immediate danger, call **911** or go to your nearest emergency department right now.

You do not need a diagnosis, a plan, or a “good enough” reason to reach out. Nor do you need to have read another word of this book. This applies exactly as much if your symptoms feel mild and confusing as it does if they feel severe.

If you are worried about a friend or family member, the same resources apply to you. You can call or text 988 to ask how to support someone else, or call 911 if there is immediate danger.

Everything else in this book can wait. This cannot.

Red Flags That Need Same-Day Medical Attention, Not a Supplement Plan

Most of the brain fog and mood symptoms this book addresses build gradually, over weeks or months, and are safe to explore with the terrain-based approach described here. A few patterns are different, and they deserve an emergency department, an urgent care visit, or a same-day call to a physician instead of a supplement plan or a symptom tracker. Please seek prompt medical evaluation if you or someone you love experiences any of the following.

- **Sudden or rapid confusion, memory loss, or personality change** over hours to a few days. This pattern can indicate a stroke, a brain bleed, an infection affecting the brain, medication toxicity, or a condition called normal pressure hydrocephalus, all of which need urgent evaluation, not a root-cause workup.
- **Any new focal neurologic sign:** weakness or numbness on one side of the body, slurred speech, sudden vision loss, loss of coordination, or a severe headache unlike any you have had before.
- **Psychosis** (hearing or seeing things that are not there, or fixed false beliefs) **or mania** (a dramatically decreased need for sleep, racing thoughts, grandiosity, or impulsive behavior that is out of character).
- **Any thoughts of suicide or of harming yourself or someone else.** See the crisis section above. This is always urgent, regardless of severity or duration.
- **Confusion or cognitive change accompanied by fever, a recent head injury, or a fall.**
- **A sudden, marked change in cognition in someone taking multiple medications,** especially after starting a new prescription or changing a dose.

These situations are emergencies or urgent medical evaluations. A functional or root-cause approach is not the right first step for any of them, and none of them are something to wait out or address with the material that follows.

On Psychiatric Medication: Read This Before Anything Else

If you take an antidepressant, an anti-anxiety medication, a mood stabilizer, or any other psychiatric prescription, this book will never tell you to stop, reduce, skip a dose, or change how you take it. That decision belongs to you and the physician or psychiatric provider managing your care, full stop. Nothing in the pages that follow is designed to create an opening for stopping a prescription on your own, and if any sentence in this book ever reads that way, it does not reflect what I intend and should be disregarded in favor of your prescriber's guidance.

This is worth taking seriously, not as a formality. Stopping an SSRI or SNRI antidepressant abruptly, or missing doses irregularly, can cause a real and well-documented discontinuation syndrome, with dizziness, electric-shock sensations in the head, flu-like symptoms, irritability, sleep disruption, and a return of the original depression or anxiety, sometimes more intensely than before. A 2025 meta-analysis pooling data across 35 studies found that withdrawal symptoms occur in roughly 43 percent of people who stop an antidepressant, and around 46 percent of people specifically stopping an SSRI, usually appearing within the first two weeks.¹ Reducing the dose gradually under a prescriber's supervision lowers that risk compared with stopping all at once, but it does not eliminate it.¹ Any adjustment to your prescription, including a gradual, medically supervised reduction, is a conversation to have with the person who prescribed it. It is never something to start on your own based on a book, a supplement label, a symptom quiz, or a feeling that you are ready.

None of this means medication is required forever, and it does not mean the approach in this book has nothing to offer you. It means that if a change to your prescription ever makes sense, your prescriber is the person to make that decision with. Everything that follows is written to run alongside psychiatric and medical care, not instead of it. If you are also seeing a therapist or psychiatrist, that relationship is not a placeholder until a root-cause approach "kicks in." For many people it is the most important part of the care they are receiving, and this book's job is to support it, not compete with it.

The Symptoms Nobody Quite Explains

You walk into a room and forget why. A word you have used a thousand times sits just out of reach. You read the same paragraph three times and still cannot tell anyone what it said. Meetings that used to be easy now take real effort to follow. And underneath the fog, or sometimes instead of it, there is a low mood or a low-grade anxiety that does not seem to have an obvious cause, that does not lift with a good night of sleep, and that makes the fog itself feel heavier.

If you have taken this to a doctor, you may have heard some version of “your labs are normal” or “this is probably just stress, aging, or a busy season of life.” Sometimes that is genuinely the full answer. Very often it is not the full answer, because standard bloodwork screens for a narrow set of conditions and stops there, and a fifteen-minute visit rarely has room for the kind of layered, connected questioning that brain fog and mood symptoms actually require.

This book is written for two overlapping groups of people. The first is anyone noticing brain fog, meaning word-finding trouble, difficulty concentrating, forgetfulness that feels new, or a general sense that your thinking is not as sharp as it used to be, without a diagnosis of any neurodegenerative disease. The second is anyone living with anxiety or depression who wants to understand what is happening in the body underneath the mood, as a complement to whatever mental health care they are already receiving or considering.

This book is explicitly **not** written for, and does not address, diagnosed dementia, Alzheimer’s disease, or Parkinson’s disease. Those are conditions that deserve ongoing care from a neurologist and a dementia care team, not a root-cause reframing, and if that describes your situation or a loved one’s, please treat this book as background context at most, not a plan. If your primary concern is ADHD, stimulant medication and dosing decisions belong with your prescriber, and this book covers ADHD-adjacent brain fog only where it overlaps the terrain below. If migraines, vertigo, POTS, or lingering post-viral fatigue are your main complaint, our companion guides *Unlock Your Migraines* and *Unlock Your Post-Viral Recovery* address those territories directly. If persistent physical fatigue and low energy, rather than thinking clearly or your mood, are the dominant symptom, our companion guide *Unlock Your Energy* goes deeper into mitochondrial function and the HPA axis as energy drivers; this book only covers that overlap where it bears directly on cognition and mood.

Below are nine keys. Each one explains a piece of the terrain underneath brain fog and mood symptoms, the mechanism behind it in plain language, and what to consider doing about it at an educational level. The keys build toward a staged roadmap and two practical worksheets near the end, including a checklist to bring to your next medical appointment.

Brain Fog and Low Mood Are Signals, Not Verdicts

The problem nobody mentions. Brain fog and low mood are usually treated as endpoints: something you simply have, the way you have a certain height. In a root-cause framework, they are almost always downstream signals instead, produced by one or more upstream drivers working on the nervous system at the same time. Those drivers include gut function, inflammation, blood sugar regulation, sleep quality, nutrient status, chronic stress and HPA axis output, and mitochondrial energy production. When several of these are mildly off at once, none of them may look dramatic on a single lab result, but together they can produce exactly the fog and low mood you are living with.

The mechanism, in plain terms. Your brain is metabolically the most demanding organ in your body relative to its size, and it has almost no tolerance for interruptions in nutrient supply, blood sugar stability, inflammatory signaling, or sleep-dependent maintenance. Because so many systems feed into brain function, brain fog and mood are two of the least specific symptoms in medicine: they can be the first sign of dozens of underlying issues, most of which are common and treatable, and a smaller number of which need urgent attention (see the red flags above). That lack of specificity is exactly why both get dismissed so often. A single test rarely tells the whole story, so it is tempting to conclude nothing is wrong, or that what is wrong is simply “stress” or “getting older,” when in fact several small, real, correctable things are adding up.

What to consider, at an educational level. This book has no interest in convincing you the problem is “all in your head” versus “all in your body.” Its goal is to walk through the specific, identifiable upstream drivers that commonly produce brain fog and mood symptoms, starting with the ones that are the most straightforward to test and the most satisfying to fix, because finding a genuinely reversible contributor is the single most valuable thing a root-cause workup can hand you. Key 2 starts there.

The Treatable Causes Worth Ruling Out First

The gap in a standard workup. Every item on this list is common, testable with standard labs, and genuinely correctable, often without any specialized functional testing at all. A root-cause approach is at its strongest exactly here: pointing someone toward a proper workup that finds a reversible cause of their cognitive or mood symptoms is more valuable than any supplement protocol.

B12 and folate. Vitamin B12 deficiency produces a well-documented range of neuropsychiatric symptoms, including depression, cognitive slowing, memory difficulty, and in more severe or prolonged cases, more dramatic presentations.² It is common in older adults, in anyone with reduced stomach acid or malabsorption (including long-term use of acid-reducing medications), in strict vegans and vegetarians, and after certain bariatric surgeries. Low folate status has also been linked to depression risk. A 2026 analysis of national U.S. survey data in women found the relationship was not simply “more is better”: both the lowest and the highest quartiles of red blood cell folate were associated with higher odds of depression compared with a mid-range level, an important reminder that nutrient status is about sufficiency, not maximizing a number.³ Ask for a B12 level (ideally with methylmalonic acid, a more sensitive marker) and a folate level, ideally measured in red blood cells rather than serum alone.

Thyroid function. Hypothyroidism, whether overt or subclinical, is a recognized cause of a genuine, distressing symptom complex that clinicians increasingly call “brain fog,” with fatigue, low mood, and difficulty with memory and executive function that often predates the formal diagnosis.⁴ There is a complication worth knowing about. A large 2021 analysis that pooled individual data from more than 74,000 adults across 21 cohorts found that subclinical hypothyroidism and hyperthyroidism were not associated with measurable differences in cognitive test scores, cognitive decline, or dementia risk at the population level, and it did not support routinely screening for subclinical thyroid dysfunction specifically to explain cognitive decline.⁵ Both things can be true at once: brain fog is a real and recognized experience for many people with thyroid dysfunction, and large population studies have not consistently found it on standardized cognitive testing, particularly for milder, subclinical thyroid abnormalities. A TSH and free T4 are simple, low-cost tests either way, and correcting a genuinely abnormal thyroid level is safe and often helps, even where the population-level cognitive testing data is more mixed than commonly assumed.

Obstructive sleep apnea. Loud snoring, witnessed pauses in breathing, gasping awake, or persistent daytime sleepiness deserve a real conversation about a sleep study, not just a comment about needing more rest. A 2026 study of nearly 2,800 cognitively unimpaired middle-aged adults found that self-reported obstructive sleep apnea was associated with poorer memory and a higher composite dementia-risk score, regardless of genetic risk status.⁶ Sleep apnea is also directly treatable, which makes it one of the higher-value items on this list to rule out.

Anemia. Low hemoglobin reduces the oxygen available to your brain, and the relationship to cognition is measurable even at levels well above what most people consider dangerously low. A 2026 study of

hospitalized older adults found that each 1 g/dL drop in hemoglobin was associated with a meaningful worsening in a standard cognitive screening score, an effect that was two to three times larger in people over 65 than in younger adults.⁷ A basic complete blood count and ferritin level will catch most cases.

Vitamin D and other B vitamins. Vitamin D deficiency and low intake of B vitamins including thiamin, riboflavin, niacin, and folate have each been linked to depressive symptoms, though a 2024 review of the evidence was candid that study designs vary widely and vitamin D supplementation trials for mood specifically have produced inconsistent results, which puts the fair summary at “worth checking and correcting if low,” not “guaranteed to fix your mood.”⁸ A vitamin D (25-OH) level is inexpensive and worth including with any of the labs above.

Medications, including over-the-counter ones. Many common medications, especially those with anticholinergic activity (some allergy medications, some bladder medications, some older antidepressants, some sleep aids), are associated with measurable cognitive impairment, particularly as the total anticholinergic burden across several medications adds up. A 2026 study using a validated anticholinergic burden scale found that people with a high anticholinergic burden had meaningfully more cognitive impairment than those with a low burden.⁹ Bring your full medication and supplement list, including anything over-the-counter, to your next appointment and ask specifically about anticholinergic burden and interactions.

Hearing. This one surprises most people. The 2020 Lancet Commission on dementia prevention identified hearing loss in midlife as one of the most significant potentially modifiable risk factors across the life course.¹⁰ A 2023 analysis using nationally representative U.S. data quantified this directly: among adults with moderate or greater hearing loss measured by audiometry, an estimated 17 percent of dementia cases were attributable to that hearing loss, roughly eight times higher than estimates based on self-reported hearing problems alone.¹¹ A hearing screen is quick, painless, and one of the more overlooked items on this entire list, especially if you are over 50 or have worked around consistent loud noise.

Putting the list to work, at an educational level. None of these eight items requires specialized functional testing to check. Bring the worksheet near the end of this book to your primary care visit and ask, specifically, for each one to be addressed rather than assumed normal. Finding one genuinely treatable cause on this list is often worth more than months of supplements aimed at a problem that was never the real driver.

A Note on Popular Mood Supplements

Two over-the-counter supplements come up constantly in conversations about mood: St. John's Wort and 5-hydroxytryptophan (5-HTP). Both have real, studied effects on serotonin activity, which is exactly why both carry a serious interaction risk that deserves more attention than it usually gets. St. John's Wort has documented, clinically significant interactions with SSRIs and other antidepressants, oral contraceptives, blood thinners, and other prescription medications, including a risk of serotonin syndrome when combined with serotonergic antidepressants.¹² 5-HTP raises serotonin more directly, and combining it with an SSRI, SNRI, or MAOI can trigger serotonin syndrome, a potentially serious reaction involving agitation, rapid heart

rate, tremor, and in severe cases, a medical emergency.¹³ Neither of these is a reason to panic if you have taken either one, but neither should be started, combined with a psychiatric prescription, or continued alongside one, without first talking to your prescriber. If you are not currently on a psychiatric medication and are considering either supplement, that conversation still belongs with a physician or my team before you start, not after.

The Gut-Brain Axis, and What Your Neurotransmitter Test Is Really Measuring

Where the science holds and the marketing runs ahead. The gut-brain axis is genuinely one of the more interesting and better-supported terrain concepts in this field, and it is also one of the most overstated in casual conversation and in some functional medicine marketing. A 2026 review describes the gut-brain axis as an integrative framework connecting gut microbial ecology, immune signaling, endocrine function, and mood regulation in depressive disorders, with mechanisms spanning short-chain fatty acid production, gut barrier integrity, and inflammatory signaling.¹⁴ That is real science. It is also a genuinely complex, still-developing field, not a settled explanation for any one person's symptoms.

What is actually happening in the body. Your gut lining, when healthy, is selective about what it lets into your bloodstream. When that barrier is compromised, a state sometimes loosely called “leaky gut,” bacterial byproducts and inflammatory signals can reach systemic circulation more easily, and systemic inflammation has real, documented effects on mood, which Key 5 covers in more depth. Your gut also produces the large majority of your body's serotonin, in the enterochromaffin cells lining your intestines, and hosts trillions of bacteria that interact with nutrient absorption, inflammation, and the vagus nerve, the primary communication line between your gut and your brain.

The caveat that comes with neurotransmitter testing. If you have had, or are considering, a urinary neurotransmitter test through a functional medicine practice, here is the straightforward picture. These panels measure real chemicals in a real sample, and the lab work itself is not the issue. The interpretation is where claims tend to run ahead of the evidence. Most of the serotonin your body makes is produced in your gut, and serotonin does not meaningfully cross the blood-brain barrier in either direction under normal conditions. The dopamine, norepinephrine, and epinephrine measured in a urine sample come overwhelmingly from your adrenal glands, your gut, and peripheral nerve activity, not from inside your skull. A detailed 2011 scientific review that specifically examined whether urinary neurotransmitter measurements reflect nervous system activity concluded that the strongest, most clinically established use for this kind of testing is detecting a rare adrenal tumor called pheochromocytoma, a use that has nothing to do with mood or cognition, and that the evidence connecting urinary neurotransmitter levels to depression, ADHD, or brain fog specifically was, in that review's own words, early and limited rather than settled.¹⁵ None of this makes the test worthless. A number on a urine panel is one data point, collected outside your brain, and it should never be handed to you as if it were a direct measurement of your brain chemistry, or used as the sole basis for a diagnosis or a treatment plan.

How to use this, at an educational level. If gut symptoms (bloating, irregular bowel habits, reflux, food reactions) travel alongside your brain fog or mood symptoms, that pattern is worth mentioning to your practitioner specifically, because gut-focused evaluation may be genuinely relevant terrain. If you are offered

a urinary neurotransmitter panel, ask what specific decision it will change, and hold its results alongside your actual symptoms and your other labs rather than as a stand-alone verdict on your brain chemistry.

Blood Sugar and the Aging Brain

The overlooked link between blood sugar and thinking. Blood sugar swings and insulin resistance are usually framed as a diabetes conversation rather than a brain conversation, yet your brain is exquisitely sensitive to both. Type 2 diabetes is increasingly recognized as a major risk factor for cognitive decline that extends well beyond its traditional metabolic complications.

The biology, without the jargon. A 2026 review describes cognitive decline associated with type 2 diabetes as arising from a convergence of gut dysbiosis, chronic low-grade inflammation, impaired insulin signaling in the brain itself, blood-brain barrier dysfunction, and mitochondrial abnormalities, a network the authors call the gut-brain-mitochondria axis.¹⁶ Insulin does more in the brain than regulate blood sugar. It participates directly in synaptic signaling and neuronal survival pathways, and when insulin signaling is chronically impaired, whether from prediabetes, insulin resistance, or diabetes itself, brain cells can become functionally starved for energy even when overall blood sugar looks acceptable on a single fasting test.

How sugar damage physically reaches the brain. There is one mechanism worth knowing by name, because it turns a vague association into something concrete. When blood sugar runs high, sugars bind to proteins and fats in a process called glycation, and the products of that reaction are called advanced glycation end products, or AGEs. AGEs act on a receptor called RAGE, the receptor for AGEs, and RAGE is the point where a metabolic problem becomes a neurological one. A 2020 review describes RAGE in Alzheimer's disease as both an intermediary of inflammation and an inducer of oxidative stress, involved in amyloid-beta production and clearance, in synaptic impairment, and in the two-way traffic between inflammation in the body and inflammation in the brain.¹⁷ The useful takeaway is not a product to buy; it is that the glycation load produced by years of high blood sugar has a well-studied route into brain tissue, which is why blood sugar keeps turning up in brain research that has nothing to do with diabetes care.

A term you are going to hear. Somewhere in this territory you will run into “type 3 diabetes” used as a nickname for Alzheimer's disease. It comes from a 2008 review that gathered the evidence for impaired insulin signaling in the Alzheimer's brain and argued that the pattern was distinct enough to deserve its own name.¹⁸ The observation underneath the nickname is genuine, and it is the same one described above. What the nickname is not is a diagnosis. No such condition appears on any diagnostic list, so no physician will ever diagnose you with it. The version that circulates online, that Alzheimer's simply is diabetes of the brain and that cutting carbohydrates therefore prevents it, claims considerably more than the research supports. Treat it as shorthand for a real line of investigation rather than as a settled answer, and be careful with anyone selling you a protocol on the strength of the nickname alone.

Where to start, at an educational level. Fasting glucose, hemoglobin A1c, and fasting insulin are worth adding to the labs from Key 2 if blood sugar has not been checked recently, particularly if brain fog is worse after meals or improves with regular eating. Beyond testing, the interventions with the most established evidence for stabilizing blood sugar (regular meal timing, adequate protein and fiber, and the movement

described in Key 8) are the same interventions that show up repeatedly throughout this book, which is exactly the point: a small number of terrain-level changes tend to move several of these dials at once.

Inflammation, the Quiet Driver of Mood Symptoms

The third contributor most conversations skip. Depression is still commonly discussed as a purely chemical or purely psychological problem. There is a substantial and growing body of research pointing to a third contributor: systemic inflammation.

How inflammation reaches mood, in plain language. A landmark 2016 review in Nature Reviews Immunology laid out the evidence that inflammatory cytokines, the signaling molecules your immune system uses to coordinate a response to infection or injury, can directly produce depressive symptoms when chronically elevated, through effects on neurotransmitter metabolism, neuroendocrine function, and neural circuits involved in reward and motivation.¹⁹ Two claims this evidence does not make: that all depression is inflammatory, and that anti-inflammatory supplements replace antidepressants or therapy. What it does say is that chronic, low-grade inflammation, whether from gut permeability, poor sleep, chronic stress, obesity, or an autoimmune process, is a genuine, biologically plausible contributor to mood symptoms in at least a meaningful subset of people, and it helps explain why addressing terrain-level inflammation sometimes improves mood symptoms that have not responded fully to standard treatment alone.

What you can do with this, at an educational level. A high-sensitivity CRP level is a simple, inexpensive screening marker for systemic inflammation. It is nonspecific: it tells you whether your body is carrying an inflammatory load, not where the load is coming from. That makes it a starting question rather than an answer, and it is worth discussing with your practitioner if fatigue, mood symptoms, and brain fog have traveled together for you. The nutrition and movement patterns in Key 8, along with the sleep and stress content in Keys 6 and 7, are the best-supported levers for lowering chronic inflammation without medication.

Sleep, Your Brain's Nightly Cleaning Shift

A cause disguised as a symptom. Poor sleep is treated as a symptom of brain fog and low mood. It is at least as accurate to treat it as a cause, because of a specific, physical cleaning process that depends on sleep to run properly.

The nightly cleaning process, explained simply. A landmark 2013 study in mice, published in the journal *Science*, discovered a system now called the glymphatic system: a network that clears metabolic waste, including proteins linked to neurodegeneration, out of brain tissue. The researchers found that natural sleep was associated with a striking increase in the space between brain cells, allowing fluid to flow through and flush out waste products far more efficiently than during wakefulness.²⁰ That finding was in mice, and the exact scale of the effect in humans is still an active area of research, but the core concept, that sleep is doing active maintenance work in your brain and not simply “resting,” is now well established and helps explain why poor sleep so reliably produces next-day brain fog, regardless of what else is going on in your health.

Practical steps, at an educational level. Consistent sleep and wake times, a dark and cool bedroom, and limiting alcohol close to bedtime (alcohol fragments sleep architecture even when it helps you fall asleep faster) are the highest-yield, lowest-cost changes here. If you snore heavily, wake gasping, or feel unrefreshed no matter how many hours you log, revisit the sleep apnea section in Key 2 rather than assuming the fix is simply more time in bed.

Chronic Stress and the HPA Axis

Why “stress” gets dismissed, and should not be. “Stress” is such a broad, overused word that it has become easy to dismiss as an explanation for brain fog and mood symptoms. The biology underneath chronic stress is specific, measurable, and directly relevant to both.

What chronic stress does physically. Your hypothalamic-pituitary-adrenal (HPA) axis is the signaling chain that governs your cortisol response to stress. A 2025 review describes how prolonged activation of the HPA axis, from chronic (not occasional) stress, disrupts normal cortisol regulation and interacts with inflammatory pathways to produce oxidative stress and structural and functional changes in the hippocampus, the brain region most central to memory formation and emotional regulation.²¹ Nobody here is saying stress is “all in your head,” or that willpower can simply override it. The research describes a real, physical feedback loop: chronic stress changes hippocampal function, and a dysregulated hippocampus makes it harder to regulate the stress response, which can compound both memory difficulty and mood symptoms over time if the loop is not interrupted.

Breaking the loop, at an educational level. Interrupting a chronic HPA axis pattern is rarely about eliminating stress, which usually is not realistic, and much more about building in genuine nervous-system downtime through consistent sleep (Key 6), regular movement (Key 8), and practices such as slow breathing, time outdoors, or meditation that have research support for lowering cortisol reactivity. If stress feels unmanageable or is tied to a specific trauma, a therapist is often the more direct and better-supported option here, not a lesser one than a supplement protocol, and pairs well with everything else in this book rather than competing with it.

Nutrition and Movement, the Two Levers With the Strongest Evidence

The least exciting advice, and the best supported. Of everything in this book, the interventions with the most rigorous supporting evidence are also the least exciting to hear about: what you eat on a normal Tuesday, and whether you move your body regularly. Neither requires a lab test or a supplement, and both are frequently the first thing to slip when brain fog and low mood make everything feel harder.

What the trials actually found. A large 2015 cohort study tracking older adults over several years found that people whose diets most closely followed the MIND diet, a hybrid of the Mediterranean and DASH dietary patterns emphasizing vegetables (especially leafy greens), berries, nuts, whole grains, fish, and olive oil, showed a substantially slower rate of cognitive decline than those whose diets least resembled it, a difference in decline rate the researchers described as equivalent to being 7.5 years younger.²² That was an observational study, not a controlled trial, so it shows a strong association rather than definitive proof, and the researchers themselves noted that a dietary intervention trial would be needed to confirm the finding, but the size and consistency of the association make it one of the better-supported nutrition patterns in this entire field. On the mood side, a randomized controlled trial called SMILES tested whether a structured dietary improvement program, delivered by a clinical dietitian alongside participants' existing therapy and medication, could improve depression symptoms. It found a significantly greater improvement in depression scores in the dietary support group compared with a social support control group, with roughly one in three participants in the dietary group reaching remission compared to one in twelve in the control group.²³ This was tested as an addition to standard care, not a replacement for it, which is exactly how to think about nutrition's role in this book.

On movement, a randomized controlled trial of 120 older adults found that a year of aerobic exercise training increased the volume of the hippocampus by about 2 percent, offsetting the equivalent of one to two years of the age-related shrinkage that region normally undergoes, alongside measurable improvements in memory.²⁴ And in the most rigorous trial to date testing a combined lifestyle approach, the FINGER trial randomly assigned older adults already at elevated dementia risk to either a two-year multidomain program (diet, exercise, cognitive training, and vascular risk monitoring) or general health advice. The intervention group showed a modest but statistically significant improvement in a composite cognitive test score compared with the control group.²⁵ The effect was real but modest, this was a "proof-of-concept" trial in older adults already flagged as at-risk, and it does not mean cognitive decline or dementia can be reliably prevented or undone in every person. Still, a combined approach, tested rigorously, measurably moved a hard outcome in a real trial, which is a meaningfully stronger claim than any single supplement or diet trend can currently make.

The order that works, at an educational level. If you take one thing from this entire book, let it be this: a genuinely Mediterranean-leaning or MIND-style eating pattern and regular aerobic movement, three or more

times weekly, are the two changes with the deepest and most consistent evidence behind them for both brain fog and mood, and they are also the two most commonly skipped in favor of a more targeted-sounding supplement. Layering supplements on top of a poor foundation is a common and expensive mistake. Building the foundation first, then adding anything else on top of it, is the order that the evidence actually supports.

Chronic Infection and Toxicant Exposure, and the Longer-Horizon Picture

The longer-horizon question readers eventually ask. Everything so far in this book addresses brain fog and mood you are noticing now. This chapter turns to a different, longer-horizon question that a lot of readers eventually ask anyway, often after watching a parent's memory decline or after reading something online: does anything I am exposed to now affect my risk many years from now? A genuine, mainstream, still-developing body of research exists on modifiable contributors to long-term cognitive risk, and it deserves an honest answer rather than either dismissal or overclaiming. One boundary bears restating before going further: this book remains written for brain fog and mood, not for diagnosed dementia, Alzheimer's disease, or Parkinson's disease, and no item below carries a claim that addressing it will prevent or reverse a neurodegenerative disease. What follows is a longer list of terrain worth understanding and, where reasonable, addressing, in the same spirit as the gut function, blood sugar, sleep, and stress content already covered.

Chronic infection. The idea that infections contribute to Alzheimer's risk has moved from a fringe theory to genuinely active, mainstream research in the last few years. A 2025 systematic review and meta-analysis pooling 26 studies and more than 1.2 million participants found herpes simplex virus (HSV) infection associated with a meaningfully higher risk of Alzheimer's disease: about 32 percent higher in case-control studies and 20 percent higher in cohort studies, with HSV-1 specifically associated with 46 percent higher odds.²⁶ A separate 2026 study following more than 8,500 participants in the United Kingdom for up to 24 years tested three infections directly, *H. pylori*, HSV-1, and varicella-zoster virus (the virus behind chickenpox and shingles), and found a significant association with dementia risk for *H. pylori* specifically (24 percent higher risk), while HSV-1 and varicella-zoster did not reach statistical significance in this particular cohort.²⁷ Different studies finding different infections significant is normal in an active research area, not a sign the whole field is unreliable, which is exactly why this book cites more than one study rather than leaning on a single result. On the treatment side, a large 2026 retrospective study of more than 230,000 matched patients found antiviral treatment for oral or facial HSV infection associated with about 13 percent lower risk of Alzheimer's disease and 17 percent lower dementia risk, though the effect was not seen for genital HSV, and the authors themselves call for prospective confirmation rather than treating the finding as settled.²⁸

Lyme disease specifically, the tick-borne infection Dr. Bredesen and others have pointed to as a contributor worth investigating, has a thinner and more mixed evidence base than the herpesvirus research above. A small 2019 case-control study in Mexico compared 108 cognitively healthy older adults with 38 people with Alzheimer's and 39 with mild cognitive impairment, and found *Borrelia burgdorferi* (the bacterium that causes Lyme disease) antibodies in 29 percent of the Alzheimer's group compared with 10 percent of healthy controls, a significant association after adjustment.²⁹ A larger, more recent 2026 systematic review pooling studies on spirochetal infections, the bacterial family that includes Lyme disease, syphilis, and leptospirosis,

found the picture for Lyme disease specifically to be heterogeneous and inconclusive, in contrast to more consistent findings for syphilis and leptospirosis.³⁰ Where the evidence nets out: chronic infection as a category, and herpesviruses and H. pylori specifically, has real and mounting support behind it. Lyme disease's specific role is a reasonable question worth further research, but is not yet established the way the herpesvirus findings are. If ongoing or past infection, including Lyme disease, Epstein-Barr virus, or a lingering post-viral pattern, feels like it fits your own history, our companion guide *Unlock Your Post-Viral Recovery* goes considerably deeper into immune activation and chronic infection than this book does.

Toxicant and environmental burden. This is a category where honesty about exposure level matters more than almost anywhere else in this book. Heavy metal neurotoxicity is well documented, but the strongest evidence comes from occupational or unusually high-dose exposure, not typical background exposure. A 2020 review of manganese, an essential nutrient at normal dietary levels, found that industrial and occupational overexposure (welding and mining are the classic examples) is linked to oxidative stress, neuroinflammation, and disrupted brain transporter function, a very different exposure picture than getting manganese from food.³¹ Mercury shows a similar pattern. A 2021 review of Amazon river communities living near artisanal gold mining found measurable cognitive, motor, and emotional deficits in populations with mercury body burden two to ten times recommended limits, while a separate review focused on dental personnel, people occupationally exposed to mercury vapor while handling amalgam, found associations with markers of neurotoxicity.³²³³ Both describe meaningfully elevated, ongoing occupational or environmental exposure, not the low-level exposure most readers have from a filling or an occasional meal of fish, and this book will not imply otherwise. If you have a specific occupational exposure history (mining, welding, dental work, or other industrial work) or live near a known contamination source, that history is worth mentioning to your practitioner directly.

Mold and mycotoxin exposure is a related concern that comes up often in functional medicine conversations, including Dr. Bredesen's. A 2018 review describes reported effects of mold and mycotoxin exposure across multiple body systems, most consistently respiratory and allergic symptoms, with nervous-system effects described as a real but considerably less developed area of research.³⁴ A specific search for evidence connecting mold exposure to long-term dementia risk did not turn up a human evidence base comparable to what exists for the infections above, and this book will not overstate a connection the research does not yet support.

Air pollution has the strongest and most consistent evidence of any environmental exposure in this chapter. A 2025 systematic review and meta-analysis pooling data across tens of millions of participants found long-term exposure to fine particulate matter (PM2.5) associated with about 8 percent higher dementia risk per 5 micrograms per cubic meter of average exposure, with positive associations also found for nitrogen dioxide and black carbon exposure, while some other pollutants (nitrogen oxides, PM10, ozone) showed no clear association.³⁵ These are modest, population-level effect sizes rather than a claim that your personal air quality determines your cognitive future. Still, they are consistent enough across a large evidence base that air pollution is one of the named factors in the Lancet Commission framework below. If reducing ongoing toxicant exposure interests you, our companion guide *Unlock Your Detox Capacity* covers this in depth,

including the honest distinction between validated, unprovoked testing and the provoked or chelation-based approaches I do not use.

Head injury. A history of significant head injury, even a single moderate or severe traumatic brain injury, is a recognized contributor to later dementia risk, and it deserves brief, honest mention even though little can be done retroactively about an injury that already happened. A 2023 meta-analysis combining civilian and military data found a history of traumatic brain injury associated with about 66 percent higher risk of later dementia, an association that did not differ meaningfully by injury severity or between veterans and civilians, though veterans as a group carry a larger population-level impact simply because head injury is more common in that population.³⁶ If you have a history of concussion, a more serious head injury, or repeated head impacts from contact sports or another cause, that history is worth mentioning to your practitioner as part of your overall picture, not as a cause for alarm on its own.

The mainstream umbrella: the Lancet Commission. Everything above can read like a scattered list of separate worries. The most authoritative attempt to organize modifiable dementia risk factors into one evidence-based framework is the Lancet Commission on dementia prevention, intervention, and care, an international panel of dementia researchers whose most recent update was published in August 2024 in *The Lancet*, following the widely cited 2020 report already referenced in Key 2 of this book.³⁷ Its named modifiable risk factors include hearing loss (Key 2 above), physical inactivity and diet (Key 8 above), and the air pollution and traumatic brain injury just discussed, among others spanning early life, midlife, and later life. Two things are worth saying plainly. First, this is genuinely mainstream, non-fringe work, published in one of medicine's most rigorous journals and reviewed by dozens of dementia specialists worldwide, not an alternative or functional medicine framework. Second, the Commission's own position is that a substantial share, not all, of dementia risk is potentially modifiable, and addressing every factor on its list does not guarantee any individual outcome. This book leans on the Commission because it is the legitimate, thoroughly vetted version of the same instinct that draws people toward more speculative frameworks: the idea that cognitive risk over a lifetime is not fixed at birth and has real, addressable contributors.

A Note on Dr. Bredesen's Multifactorial Approach

If you have read about Dr. Dale Bredesen's work, this is a fair place to address it directly, since the terrain covered in this chapter overlaps with territory he has written about for years. Dr. Bredesen is a neurologist and researcher affiliated with UCLA's David Geffen School of Medicine who has argued that cognitive decline usually has multiple simultaneous contributors, including chronic infection and toxicant exposure among others, rather than one single cause, and that a personalized workup addressing several contributors at once may help more than treating cognitive decline as a single disease process. That underlying instinct, that cognitive decline is often multifactorial and that infections and toxicants belong on the list of things worth investigating, is consistent with the mainstream research already described above. Where this book draws a firm line is around his specific published clinical evidence to date. The primary published trial of his personalized, multi-domain protocol enrolled 25 patients with dementia or mild cognitive impairment in a single-arm design with no control group, described by its own authors as a proof-of-concept study, and its own authors explicitly call for a larger randomized controlled trial before the approach can be considered

established.³⁸ That is an honest, preliminary starting point for further research, not the same thing as a proven protocol. This book does not adopt his specific diagnostic framework or terminology. What it does share with his work is the well-supported underlying idea that chronic infection and toxicant exposure belong on the list of things worth understanding about long-term brain health, which is exactly why they appear above.

A Note on Statins and Brain Health

Some clinicians, Dr. Bredesen among them, advise avoiding statin medications specifically for brain health reasons. That advice deserves a careful look, not dismissal and not alarm. In February 2012, the FDA did add a label change to statin medications noting rare post-marketing reports of cognitive effects, memory loss, forgetfulness, and confusion, generally reversible within weeks of stopping the medication.³⁹ That label change happened, and it is worth knowing about. But it does not mean statins are, on balance, harmful to the brain. A large 2022 meta-analysis pooling 36 observational studies found statin use associated with about 20 percent lower odds of dementia and 32 percent lower odds for Alzheimer's disease specifically, with no meaningful difference between lipophilic statins (which cross the blood-brain barrier more readily) and hydrophilic statins (which do not), a finding that directly undercuts the theory that statins harm cognition by crossing into the brain.⁴⁰ A separate 2021 study following nearly 19,000 older adults for almost five years found no association between statin use and incident dementia, mild cognitive impairment, or cognitive decline.⁴¹ Memory complaints are a real, labeled, reversible effect reported by a small number of people taking statins, but they are not a common side effect, and the weight of current evidence does not show statins increasing dementia risk. As with every medication decision in this book, the answer is never to stop or reduce a statin on your own. If you are concerned about a statin's effect on your memory or thinking, that is a conversation to have with the physician who prescribed it, exactly the way this book approaches every other prescription decision.

How to hold all of this, at an educational level. None of this changes the boundary stated at the start of this book. It remains written for brain fog and mood, not for diagnosed dementia, Alzheimer's disease, or Parkinson's disease. Addressing any of these contributors comes with no promise of preventing or reversing a neurodegenerative disease in you or anyone else. What this chapter offers is an honest, mainstream-grounded answer to a question many readers eventually ask: if you are thinking about your brain health over decades, not just this week, what is actually worth knowing about? Reducing ongoing toxicant exposure where that is reasonable, treating a genuine chronic infection if one is found, protecting your head during activities with real injury risk, and knowing your own occupational or environmental exposure history are all reasonable, low-risk things to know and to mention to your practitioner. None of them replace the higher-certainty, more immediately actionable items in Key 2, and none of them are a reason to start an expensive testing protocol without a specific reason to suspect it applies to you.

A Staged Roadmap

The first two weeks. Bookmark the crisis resources at the front of this book. Start the symptom and sleep tracker below so you have a real baseline, not a memory of one. Schedule an appointment with your primary care provider or my team and bring the treatable-causes checklist below. If you are on a psychiatric medication, change nothing about it on your own.

Weeks three through eight. Begin shifting your eating pattern toward the MIND-style approach in Key 8, and add regular aerobic movement if it is not already part of your routine, starting wherever your current fitness allows. Address whatever came back on your labs from Key 2. Correct a real deficiency, treat a real thyroid abnormality, start sleep apnea evaluation if indicated, and review your medication list for anticholinergic burden. Tighten sleep and wake times per Key 6.

Ongoing. Continue tracking. If brain fog or mood symptoms are not meaningfully better after 8 to 12 weeks of addressing the treatable causes and building the nutrition and movement foundation, that is useful information, not a failure, and it is the point at which more specialized testing (gut-focused, inflammatory, or otherwise) becomes genuinely worth considering with a practitioner, rather than being the first move.

Your Symptom and Sleep Tracker

Rate each item once daily for at least two weeks before making any change, so you have a real baseline to compare against. Repeat the same tracker after 30 and 90 days of any new approach to see whether it actually helped, since memory alone is a notoriously unreliable way to judge gradual change.

Date	Hours slept	Sleep quality (1 to 5)	Brain fog (1 to 10)	Mood (1 to 10)	Stress (1 to 10)	Caffeine after 2pm (Y/N)	Alcohol (Y/N)	Movement today (Y/N)	Notes

Your Treatable-Causes Checklist

Bring this list to your next medical appointment and ask that each item be specifically addressed rather than assumed normal.

- ☐ Vitamin B12, and ideally methylmalonic acid (MMA), a more sensitive marker
- ☐ Folate, ideally red blood cell folate rather than serum alone
- ☐ TSH and free T4 (and free T3 if your practitioner agrees it is warranted)
- ☐ Complete blood count (CBC) and ferritin
- ☐ Vitamin D (25-OH)
- ☐ A full medication and supplement list, reviewed specifically for anticholinergic burden and interactions
- ☐ A conversation about snoring, witnessed breathing pauses, or daytime sleepiness, with a referral for a sleep study if indicated
- ☐ A hearing screen, especially if you are over 50 or have significant noise exposure history
- ☐ Fasting glucose, hemoglobin A1c, or fasting insulin, if blood sugar has not been checked recently
- ☐ High-sensitivity CRP, if inflammation has not been assessed and mood, fatigue, and brain fog are traveling together

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a complimentary consultation with my team to review your specific symptoms, which of the treatable causes in this book have and have not been checked, and whether a more personalized workup makes sense for your situation. There is no obligation, and everything in this conversation is meant to run alongside, not replace, any mental health or medical care you are already receiving.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Not ready for a conversation yet? Start tracking.

The tracker earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you the connections that are hard to see from inside them: how sleep tracks against next-day clarity, and whether mood and focus move together or separately. Two weeks of honest data makes any future appointment, with our team, your physician, or your mental health provider, dramatically more useful than memory alone.

Start your free tracker at rootcauseunlocked.com/tracker.

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If you are having thoughts of suicide or self-harm, please call or text 988, call 911, or go to your nearest emergency department before doing anything else.

This guide is provided for general educational purposes and does not constitute individualized medical, psychiatric, or psychological advice, diagnosis, or treatment, and it does not diagnose dementia, Alzheimer's disease, depression, anxiety, or any other condition. It is not a substitute for a relationship with your physician, psychiatrist, therapist, or other qualified health care provider. If you are having thoughts of suicide or self-harm, call or text 988, call 911, or go to your nearest emergency department immediately. Never stop, start, reduce, or otherwise change the dose of any prescription medication, including antidepressants, anti-anxiety medications, or mood stabilizers, without first talking to the prescriber managing your care. Always consult your physician before starting any new supplement, particularly if you are pregnant, nursing, taking prescription medications, or being treated for a psychiatric condition.

References

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1. Zhang MM, Tan X, Zheng YB, Zeng N, Li Z, Horowitz MA, Feng XZ, Wang K, Li ZY, Zhu WL, Zhou X, Xie P, Zhang X, Wang Y, Shi J, Bao YP, Lu L, Li SX. Incidence and risk factors of antidepressant withdrawal symptoms: a meta-analysis and systematic review. *Molecular Psychiatry*. 2025 May;30(5):1758-1769. PMID 39394455. <https://pubmed.ncbi.nlm.nih.gov/39394455/>
2. Sahu P, Thippeswamy H, Chaturvedi SK. Neuropsychiatric manifestations in vitamin B12 deficiency. *Vitamins and Hormones*. 2022;119:457-470. PMID 35337631. <https://pubmed.ncbi.nlm.nih.gov/35337631/>
3. Zeng W, Ma H, Wei S, Li H, Jin Z, Zhou L, Chai S, Tan Y, Tang G, Wei H. Red blood cell folate and depression risk in U.S. women: A dose-response analysis from a nationwide cross-sectional study (2009-2018). *Medicine (Baltimore)*. 2026 Mar 13;105(11):e48023. PMID 41824854. <https://pubmed.ncbi.nlm.nih.gov/41824854/>
4. Samuels MH, Bernstein LJ. Brain Fog in Hypothyroidism: What Is It, How Is It Measured, and What Can Be Done About It. *Thyroid*. 2022 Jul;32(7):752-763. PMID 35414261. <https://pubmed.ncbi.nlm.nih.gov/35414261/>
5. van Vliet NA, van Heemst D, Almeida OP, Asvold BO, Aubert CE, Bae JB, Barnes LE, Bauer DC, Blauw GJ, Brayne C, Cappola AR, Ceresini G, Comijs HC, Dartigues JF, Degryse JM, Dullaart RPF, van Eersel MEA, den Elzen WPJ, Ferrucci L, Fink HA, Flicker L, Grabe HJ, Han JW, Helmer C, Huisman M, Ikram MA, Imaizumi M, de Jongh RT, Jukema JW, Kim KW, Kuller LH, Lopez OL, Mooijaart SP, Moon JH, Moutzouri E, Nauck M, Parle J, Peeters RP, Samuels MH, Schmidt CO, Schminke U, Slagboom PE, Stordal E, Vaes B, Volzke H, Westendorp RGJ, Yamada M, Yeap BB, Rodondi N, Gussekloo J, Trompet S. Association of Thyroid Dysfunction With Cognitive Function: An Individual Participant Data Analysis. *JAMA Internal Medicine*. 2021 Nov 1;181(11):1440-1450. PMID 34491268. <https://pubmed.ncbi.nlm.nih.gov/34491268/>
6. Abdelmessih GT, Bransby L, Cummins H, Jackson ML, Lim YY. Associations of self-reported obstructive sleep apnea with cognition and dementia risk in cognitively unimpaired middle-aged adults. *Alzheimer's & Dementia*. 2026 Jul;22(7):e71553. PMID 42370433. <https://pubmed.ncbi.nlm.nih.gov/42370433/>
7. Prochaska M, Harper E. The Association Between Hemoglobin Level and Cognitive Function in Older Adults During Hospitalization: A Prospective Observational Study. *Health Science Reports*. 2026 Jul 25;9(8):e72857. PMID 42504232. <https://pubmed.ncbi.nlm.nih.gov/42504232/>
8. Jahan-Mihan A, Stevens P, Medero-Alfonso S, Brace G, Overby LK, Berg K, Labyak C. The Role of Water-Soluble Vitamins and Vitamin D in Prevention and Treatment of Depression and Seasonal Affective Disorder in Adults. *Nutrients*. 2024 Jun 17;16(12):1902. PMID 38931257. <https://pubmed.ncbi.nlm.nih.gov/38931257/>
9. Ruiz-Julian M, Piñol-Ripoll G. Anticholinergic burden and its relationship with cognitive impairment in a neuropsychiatric unit. *Farmacia Hospitalaria*. 2026 Jul-Aug;50(4):T247-T252. PMID 42156261. <https://pubmed.ncbi.nlm.nih.gov/42156261/>
10. Livingston G, Huntley J, Sommerlad A, Ames D, Ballard C, Banerjee S, Brayne C, Burns A, Cohen-Mansfield J, Cooper C, Costafreda SG, Dias A, Fox N, Gitlin LN, Howard R, Kales HC, Kivimaki M, Larson EB, Ogunniyi A, Orgeta V, Ritchie K, Rockwood K, Sampson EL, Samus Q, Schneider LS, Selbaek G, Teri L, Mukadam N. Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*. 2020 Aug 8;396(10248):413-446. PMID 32738937. <https://pubmed.ncbi.nlm.nih.gov/32738937/>

11. Smith JR, Huang AR, Lin FR, Reed NS, Deal JA. The Population Attributable Fraction of Dementia From Audiometric Hearing Loss Among a Nationally Representative Sample of Community-Dwelling Older Adults. *The Journals of Gerontology Series A*. 2023 Jul 8;78(7):1300-1306. PMID 37139951. <https://pubmed.ncbi.nlm.nih.gov/37139951/>
12. Caus MN, Lupoae M, Chitescu CL. Efficacy and Safety of Herbal Supplements with Anxiolytic, Antidepressant, and Sedative Action: A Review of Clinical Data and Toxicological Risks. *Pharmaceuticals (Basel)*. 2026 Feb 28;19(3):399. PMID 41901246. <https://pubmed.ncbi.nlm.nih.gov/41901246/>
13. Dome P, Tombor L, Lazary J, Gonda X, Rihmer Z. Natural health products, dietary minerals and over-the-counter medications as add-on therapies to antidepressants in the treatment of major depressive disorder: a review. *Brain Research Bulletin*. 2019 Mar;146:51-78. PMID 30599219. <https://pubmed.ncbi.nlm.nih.gov/30599219/>
14. Zhao H, Ao L, Hao L, Wei Y, Yin HZ, Lee XQ, Guo C, Wang Z, Yang J, Yang R, Zhou GL. Integrative mechanisms and intervention targets of the microbiota-gut-brain axis in depressive disorders: advances across immune, endocrine, and central nervous system pathways. *Frontiers in Psychiatry*. 2026 Jun 8;17:1848918. PMID 42338743. <https://pubmed.ncbi.nlm.nih.gov/42338743/>
15. Marc DT, Ailts JW, Campeau DC, Bull MJ, Olson KL. Neurotransmitters excreted in the urine as biomarkers of nervous system activity: validity and clinical applicability. *Neuroscience and Biobehavioral Reviews*. 2011 Jan;35(3):635-644. PMID 20696183. <https://pubmed.ncbi.nlm.nih.gov/20696183/>
16. Nirala N, Kuwar OK, Rana D, Dhingra MS. The gut-brain-mitochondria axis in type 2 diabetes-associated cognitive decline: mechanistic convergence, biomarkers, and therapeutic opportunities. *Acta Diabetologica*. 2026 Jul 30. PMID 42530574. <https://pubmed.ncbi.nlm.nih.gov/42530574/>
17. Chellappa RC, Palanisamy R, Swaminathan K. RAGE Isoforms, its Ligands and their Role in Pathophysiology of Alzheimer's Disease. *Current Alzheimer Research*. 2020;17(14):1262-1279. PMID 33602095. <https://pubmed.ncbi.nlm.nih.gov/33602095/>
18. de la Monte SM, Wands JR. Alzheimer's disease is type 3 diabetes-evidence reviewed. *Journal of Diabetes Science and Technology*. 2008 Nov;2(6):1101-1113. PMID 19885299. <https://pubmed.ncbi.nlm.nih.gov/19885299/>
19. Miller AH, Raison CL. The role of inflammation in depression: from evolutionary imperative to modern treatment target. *Nature Reviews Immunology*. 2016 Jan;16(1):22-34. PMID 26711676. <https://pubmed.ncbi.nlm.nih.gov/26711676/>
20. Xie L, Kang H, Xu Q, Chen MJ, Liao Y, Thiyagarajan M, O'Donnell J, Christensen DJ, Nicholson C, Iliff JJ, Takano T, Deane R, Nedergaard M. Sleep drives metabolite clearance from the adult brain. *Science*. 2013 Oct 18;342(6156):373-377. PMID 24136970. <https://pubmed.ncbi.nlm.nih.gov/24136970/>
21. Lei AA, Phang VWX, Lee YZ, Kow ASF, Tham CL, Ho YC, Lee MT. Chronic Stress-Associated Depressive Disorders: The Impact of HPA Axis Dysregulation and Neuroinflammation on the Hippocampus, A Mini Review. *International Journal of Molecular Sciences*. 2025 Mar 24;26(7):2940. PMID 40243556. <https://pubmed.ncbi.nlm.nih.gov/40243556/>
22. Morris MC, Tangney CC, Wang Y, Sacks FM, Barnes LL, Bennett DA, Aggarwal NT. MIND diet slows cognitive decline with aging. *Alzheimer's & Dementia*. 2015 Sep;11(9):1015-1022. PMID 26086182. <https://pubmed.ncbi.nlm.nih.gov/26086182/>
23. Jacka FN, O'Neil A, Opie R, Itsiopoulos C, Cotton S, Mohebbi M, Castle D, Dash S, Mihalopoulos C, Chatterton ML, Brazionis L, Dean OM, Hodge AM, Berk M. A randomised controlled trial of dietary improvement for adults with major depression (the 'SMILES' trial). *BMC Medicine*. 2017 Jan 30;15(1):23. PMID 28137247. <https://pubmed.ncbi.nlm.nih.gov/28137247/>

24. Erickson KI, Voss MW, Prakash RS, Basak C, Szabo A, Chaddock L, Kim JS, Heo S, Alves H, White SM, Wojcicki TR, Mailey E, Vieira VJ, Martin SA, Pence BD, Woods JA, McAuley E, Kramer AF. Exercise training increases size of hippocampus and improves memory. *Proceedings of the National Academy of Sciences*. 2011 Feb 15;108(7):3017-3022. PMID 21282661. <https://pubmed.ncbi.nlm.nih.gov/21282661/>
25. Ngandu T, Lehtisalo J, Solomon A, Levalahti E, Ahtiluoto S, Antikainen R, Backman L, Hanninen T, Jula A, Laatikainen T, Lindstrom J, Mangialasche F, Paajanen T, Pajala S, Peltonen M, Rauramaa R, Stigsdotter-Neely A, Strandberg T, Tuomilehto J, Soininen H, Kivipelto M. A 2 year multidomain intervention of diet, exercise, cognitive training, and vascular risk monitoring versus control to prevent cognitive decline in at-risk elderly people (FINGER): a randomised controlled trial. *Lancet*. 2015 Jun 6;385(9984):2255-2263. PMID 25771249. <https://pubmed.ncbi.nlm.nih.gov/25771249/>
26. Ji Q, Lian W, Liu W, Tang L, Hu Z, Li L, Wang Y, Tao E, Zhan Y. Herpes Simplex Virus Infection and Risk of Alzheimer's Disease: A Systematic Review and Meta-Analysis. *Neuroepidemiology*. 2025 Sep 11:1-14. PMID 40934136. <https://pubmed.ncbi.nlm.nih.gov/40934136/>
27. Littlejohns TJ, Liu W, Brayne C, Butt J, Collister JA, Forouhi NG, Gkrania-Klotsas E, Hayat S, Jeffery K, Kuzma E, Luben RN, Mentzer AJ, Parsonnet J, Wareham N, Waterboer T, Hunter DJ. *Helicobacter pylori*, herpes simplex virus-1, varicella-zoster virus, and dementia risk. *Alzheimer's & Dementia (Amsterdam)*. 2026 Jul 6;18(3):e70414. PMID 42441168. <https://pubmed.ncbi.nlm.nih.gov/42441168/>
28. Kabakova M, Patel P, Bitterman D, Wang JY, Zafar K, Fife DA, Cohen M, Kurtti A, Jagdeo J. Antiviral Treatment of Herpes Simplex Virus Decreases the Risk of Alzheimer's Disease and Dementia. *Journal of Drugs in Dermatology*. 2026 Mar 1;25(3):234-239. PMID 41779765. <https://pubmed.ncbi.nlm.nih.gov/41779765/>
29. Herrera-Landero A, Amaya-Sanchez LE, D Hyver de Las-Deses C, Solorzano-Santos F, Gordillo-Perez MG. *Borrelia burgdorferi* as a risk factor for Alzheimer's dementia and mild cognitive impairment. *European Geriatric Medicine*. 2019 Jun;10(3):493-500. PMID 34652786. <https://pubmed.ncbi.nlm.nih.gov/34652786/>
30. Horton M, Whitley DJ, Mayhew M, McLean S. Association between spirochaetal infection and neurodegenerative diseases: a systematic review and quantitative synthesis of observational studies. *Journal of Medical Microbiology*. 2026 Mar;75(3). PMID 41790564. <https://pubmed.ncbi.nlm.nih.gov/41790564/>
31. Miah MR, Ijomone OM, Okoh COA, Ijomone OK, Akingbade GT, Ke T, Krum B, da Cunha Martins A Jr, Akinyemi A, Aranoff N, Antunes Soares FA, Bowman AB, Aschner M. The effects of manganese overexposure on brain health. *Neurochemistry International*. 2020 May;135:104688. PMID 31972215. <https://pubmed.ncbi.nlm.nih.gov/31972215/>
32. Santos-Sacramento L, Arrifano GP, Lopes-Araujo A, Augusto-Oliveira M, Albuquerque-Santos R, Takeda PY, Souza-Monteiro JR, Macchi BM, do Nascimento JLM, Lima RR, Crespo-Lopez ME. Human neurotoxicity of mercury in the Amazon: A scoping review with insights and critical considerations. *Ecotoxicology and Environmental Safety*. 2021 Jan 15;208:111686. PMID 33396018. <https://pubmed.ncbi.nlm.nih.gov/33396018/>
33. Bjorklund G, Hilt B, Dadar M, Lindh U, Aaseth J. Neurotoxic effects of mercury exposure in dental personnel. *Basic & Clinical Pharmacology & Toxicology*. 2019 May;124(5):568-574. PMID 30589214. <https://pubmed.ncbi.nlm.nih.gov/30589214/>
34. Ratnaseelan AM, Tsilioni I, Theoharides TC. Effects of Mycotoxins on Neuropsychiatric Symptoms and Immune Processes. *Clinical Therapeutics*. 2018 Jun;40(6):903-917. PMID 29880330. <https://pubmed.ncbi.nlm.nih.gov/29880330/>

35. Best Rogowski CB, Bredell C, Shi Y, Tien-Smith A, Szybka M, Fung KW, Hong L, Phillips V, Jovanovic Andersen Z, Sharp SJ, Woodcock J, Brayne C, Navaratnam A, Khreis H. Long-term air pollution exposure and incident dementia: a systematic review and meta-analysis. *Lancet Planetary Health*. 2025 Jul;9(7):101266. PMID 40716448. <https://pubmed.ncbi.nlm.nih.gov/40716448/>
36. Gardner RC, Bahorik A, Kornblith ES, Allen IE, Plassman BL, Yaffe K. Systematic Review, Meta-Analysis, and Population Attributable Risk of Dementia Associated with Traumatic Brain Injury in Civilians and Veterans. *Journal of Neurotrauma*. 2023 Apr;40(7-8):620-634. PMID 36305374. <https://pubmed.ncbi.nlm.nih.gov/36305374/>
37. Livingston G, Huntley J, Liu KY, Costafreda SG, Selbaek G, Alladi S, Ames D, Banerjee S, Burns A, Brayne C, Fox NC, Ferri CP, Gitlin LN, Howard R, Kales HC, Kivimaki M, Larson EB, Nakasujja N, Rockwood K, Samus Q, Shirai K, Singh-Manoux A, Schneider LS, Walsh S, Yao Y, Sommerlad A, Mukadam N. Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. *Lancet*. 2024 Aug 10;404(10452):572-628. PMID 39096926. <https://pubmed.ncbi.nlm.nih.gov/39096926/>
38. Toups K, Hathaway A, Gordon D, Chung H, Raji C, Boyd A, Hill BD, Hausman-Cohen S, Attarha M, Chwa WJ, Jarrett M, Bredeisen DE. Precision Medicine Approach to Alzheimer's Disease: Successful Pilot Project. *Journal of Alzheimer's Disease*. 2022;88(4):1411-1421. PMID 35811518. <https://pubmed.ncbi.nlm.nih.gov/35811518/>
39. Jukema JW, Cannon CP, de Craen AJ, Westendorp RG, Trompet S. The controversies of statin therapy: weighing the evidence. *Journal of the American College of Cardiology*. 2012 Sep 4;60(10):875-881. PMID 22902202. <https://pubmed.ncbi.nlm.nih.gov/22902202/>
40. Olmastroni E, Molari G, De Beni N, Colpani O, Galimberti F, Gazzotti M, Zambon A, Catapano AL, Casula M. Statin use and risk of dementia or Alzheimer's disease: a systematic review and meta-analysis of observational studies. *European Journal of Preventive Cardiology*. 2022 May 5;29(5):804-814. PMID 34871380. <https://pubmed.ncbi.nlm.nih.gov/34871380/>
41. Zhou Z, Ryan J, Ernst ME, Zoungas S, Tonkin AM, Woods RL, McNeil JJ, Reid CM, Curtis AJ, Wolfe R, Wrigglesworth J, Shah RC, Storey E, Murray A, Orchard SG, Nelson MR, ASPREE Investigator Group. Effect of Statin Therapy on Cognitive Decline and Incident Dementia in Older Adults. *Journal of the American College of Cardiology*. 2021 Jun 29;77(25):3145-3156. PMID 34167639. <https://pubmed.ncbi.nlm.nih.gov/34167639/>

From Understanding to Doing

Everything up to this point was about understanding what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

If you are in crisis right now, if you are thinking about harming yourself or ending your life, put this guide down and call or text **988** (the Suicide and Crisis Lifeline, available 24/7). Nothing in any lab panel matters more than that call.

Same-day medical attention, not a lab plan. Sudden confusion, sudden severe headache, one-sided weakness or facial droop, trouble speaking, a first seizure, new memory loss after a head injury, or hallucinations. These are emergencies.

Nothing in this guide is a reason to stop, skip, or reduce a psychiatric medication. Stopping an antidepressant or any psychiatric prescription abruptly can cause genuine withdrawal effects and worse. Every medication decision belongs with your prescriber.

The nine keys, and what tests each one

Unlock Your Brain Health is organized around nine keys. Here is how each maps to something measurable, and where no blood test exists at all.

Key	What it means	How you find it
1. Signals, not verdicts	Brain fog and low mood are symptoms with causes	The workup below, not a self-diagnosis
2. Treatable causes first	The highest-yield chapter in the book	This entire guide
3. Gut-brain axis	Real science, often oversold	Symptoms pattern; see the honesty section below
4. Blood sugar	The brain runs on stable fuel	Fasting glucose, HbA1c, fasting insulin
5. Inflammation	A quiet driver of mood symptoms	hs-CRP
6. Sleep	The nightly cleaning shift	A sleep study if apnea signs. Not a blood test
7. Stress and the HPA axis	A physical feedback loop	History and pattern. Not a blood test
8. Nutrition and movement	The two strongest levers	The diet guide. Not a blood test
9. Longer-horizon terrain	Infections, toxicants, hearing	Targeted tests where history points

Three of the highest-value checks in this guide are not blood draws at all. They are a sleep study, a hearing screen, and a medication review. They appear below with the labs because they belong in the same workup.

The treatable-causes workup, item by item

Every item here is common, testable, and genuinely correctable. Finding one true item on this list is worth more than months of supplements aimed at a problem that was never the driver.

B12, and the two markers that catch what B12 misses

Vitamin B12 deficiency produces a well-documented range of neuropsychiatric symptoms, including depression, cognitive slowing, and memory difficulty (PMID 35337631). It is common in older adults, in long-term users of acid-reducing medications, in vegans and vegetarians, and after bariatric surgery.

The testing subtlety that matters: serum B12 does not always reflect what your cells are actually getting. Methylmalonic acid (MMA) is the most sensitive functional marker of B12 status, and homocysteine rises when B12, folate, or B6 run short. A B12 in the low-normal range with an elevated MMA is a functional deficiency that a B12-only test would have waved through. One caution from my own interpretive framework: MMA also rises when kidney function is impaired, so it is read alongside creatinine.

Folate, with a U-shaped surprise

Ask for red blood cell folate rather than serum folate; it reflects roughly three months of tissue status rather than this week's meals. And note what a national analysis in U.S. women found: both the lowest and the highest folate levels were associated with higher odds of depression compared with the middle (PMID 41824854). Nutrient status is about sufficiency, not maximizing a number, which is a sentence worth applying to everything in this guide.

Thyroid, with both halves of the truth

Hypothyroid brain fog is a genuine, recognized clinical experience, with fatigue, low mood, and executive-function complaints that often precede diagnosis (PMID 35414261). And the honest other half: an analysis pooling individual data from more than 74,000 adults found subclinical thyroid dysfunction was not associated with measurable cognitive differences at the population level (PMID 34491268). Both can be true. A TSH and free T4 are simple, inexpensive tests; correcting a genuinely abnormal result is worthwhile on its own terms. Thyroid targets are individual enough, by age, sex, and treatment status, that this guide deliberately publishes no thyroid reference table; that conversation belongs with your clinician over your actual numbers.

Blood count and iron

Low hemoglobin means less oxygen delivery to the brain, and the cognitive association is measurable well above dangerously low levels, with a larger effect in older adults (PMID 42504232). A complete blood count plus ferritin catches most of it. Ferritin is also an acute-phase reactant, so it reads alongside hs-CRP: a “normal” ferritin during inflammation can hide depleted iron.

Vitamin D

Low vitamin D has been linked to depressive symptoms, and the honest grade from the review literature is that supplementation trials for mood are inconsistent: worth checking and correcting if low, not guaranteed to fix anything (PMID 38931257).

Blood sugar, three ways

Fasting glucose, HbA1c, and fasting insulin together read the fuel-stability story from Key 4. The brain notices insulin resistance years before a diabetes diagnosis, and brain fog that is worse after meals or better with regular eating is a pattern worth bringing up specifically.

Inflammation

hs-CRP is the inexpensive rough read on systemic inflammation, which has documented links to mood symptoms in a meaningful subset of people (PMID 26711676). A single elevated reading after illness means little; repeat it when well.

Homocysteine, read correctly

Homocysteine is a methylation-cycle marker. It rises when B12, folate, or B6 run short, and with thyroid or kidney trouble. Do not let anyone wave off an elevated result as trivia. In my read this is one of the most consequential numbers in the panel, because it tells you the methylation machinery your brain chemistry depends on is running short somewhere findable. The way to act on it is not to chase the number with blind supplementation. Find which input is short, B12, folate, B6, thyroid, or kidney function, correct that, and let homocysteine confirm the fix on retest. My functional target of 5 to 7 in the table below reflects how seriously I take it. Note that it is a window, not a floor. Lower is not better: a homocysteine driven below that window points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result.

The three checks that are not blood tests

A sleep study, if the signs are there. Loud snoring, witnessed pauses, gasping awake, or unrefreshing sleep deserve a real sleep evaluation. Self-reported sleep apnea was associated with poorer memory and higher dementia-risk scores in cognitively healthy middle-aged adults (PMID 42370433), and apnea is directly treatable, which makes it one of the highest-value items in this entire guide.

A hearing screen, especially past 50. The Lancet Commission identified midlife hearing loss as one of the most significant modifiable dementia risk factors across the life course (PMID 32738937), and audiometry-based U.S. data attributed an estimated 17 percent of dementia cases among those with moderate or greater hearing loss to the hearing loss itself (PMID 37139951). A hearing screen is quick, painless, and almost never on anyone's brain fog list. Put it on yours.

A medication review, including over-the-counter. Anticholinergic burden, the cumulative load of common allergy, bladder, sleep, and some antidepressant medications, is associated with measurably more cognitive impairment at higher burden (PMID 42156261). Bring every bottle, including supplements, to one appointment and ask the specific question: "What is my total anticholinergic burden?"

The panel to be skeptical of: urinary neurotransmitter testing

If you have been offered a urinary neurotransmitter panel as a readout of your brain chemistry, here is the honest picture I owe you. The chemicals are real, and the problem lies in the interpretation rather than the lab work. Most serotonin is made in the gut, serotonin does not meaningfully cross the blood-brain barrier, and urinary dopamine, norepinephrine, and epinephrine come overwhelmingly from the adrenal glands, gut, and peripheral nerves, not from inside your skull. The scientific review that examined this question directly concluded the established clinical use of such testing is detecting a rare adrenal tumor, and that the evidence connecting urinary levels to mood or attention was early and limited (PMID 20696183). A number on a urine panel is one data point collected outside your brain. It should never be sold to you as a brain-chemistry readout or serve as the sole basis for a diagnosis or treatment plan, here or anywhere.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose. Thyroid values are deliberately absent (see above); hemoglobin ranges differ by sex, so the row below is approximate and your lab's sex-specific range governs.

Marker	Units	Functional target	Approximate standard range
Vitamin B12	pg/mL	500 to 900	200 to 1100
Methylmalonic acid (MMA)	nmol/L	under 162	under 243
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	5 to 7	under 10 to 15, varies by lab
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Hemoglobin	g/dL	13.0 to 17.5	10 to 18, sex-specific
Ferritin	ng/mL	50 to 150	25 to 300
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0

Reading your results as a pattern

Pattern one: a treatable cause found. Low B12 with high MMA, an abnormal thyroid, anemia, sleep apnea, a heavy anticholinergic list. The best outcome this guide can produce is a specific, correctable driver, addressed with your clinician, before any supplement conversation.

Pattern two: everything normal, symptoms real. Common, and not a dead end. It moves the attention to the keys that have no blood test, which are sleep, stress, movement, nutrition pattern, hearing, and the gut-brain terrain. That is where the diet guide and the daily work live.

Pattern three: the metabolic cluster. Creeping glucose, HbA1c drifting, low vitamin D, rising hs-CRP. The brain complaint and the metabolic picture are usually one story; Key 4's terrain work serves both.

Pattern four: mood symptoms dominating. Labs support the workup; they do not replace mental health care. Therapy and, where prescribed, medication are first-class treatments, and everything in this kit is designed to work alongside them, never instead of them.

The conversation with your doctor

A request list you can bring

- B12 with **methylmalonic acid**, and red blood cell folate
- TSH and free T4
- Complete blood count and ferritin
- Vitamin D (25-OH)
- Fasting glucose, HbA1c, fasting insulin
- hs-CRP
- Homocysteine, read as a B-vitamin, thyroid, and kidney workup trigger
- A medication review for anticholinergic burden, with every bottle on the table
- A sleep study referral if snoring, gasping, or unrefreshing sleep applies
- A hearing screen, especially if over 50 or noise-exposed

Your next step

The companion guides do the rest: **The Brain Health Diet Guide** covers the two levers with the strongest evidence in this territory, and **The Brain Health Supplement Guide** covers what genuinely helps, what failed its biggest trial, and the interaction risks that matter most alongside psychiatric medication.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, start with two weeks of honest tracking:

Free tracker: rootcauseunlocked.com/tracker

References

All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in [Brain-Kit-Citation-Ledger.md](#).

1. Sahu P, Thippeswamy H, Chaturvedi SK. Neuropsychiatric manifestations in vitamin B12 deficiency. *Vitam Horm*. 2022;119:457-470. PMID 35337631
2. Zeng W, Ma H, Wei S, et al. Red blood cell folate and depression risk in U.S. women: A dose-response analysis from a nationwide cross-sectional study (2009-2018). *Medicine (Baltimore)*. 2026;105(11):e48023. PMID 41824854
3. Samuels MH, Bernstein LJ. Brain Fog in Hypothyroidism: What Is It, How Is It Measured, and What Can Be Done About It. *Thyroid*. 2022;32(7):752-763. PMID 35414261
4. van Vliet NA, van Heemst D, Almeida OP, et al. Association of Thyroid Dysfunction With Cognitive Function: An Individual Participant Data Analysis. *JAMA Intern Med*. 2021;181(11):1440-1450. PMID 34491268
5. Prochaska M, Harper E. The Association Between Hemoglobin Level and Cognitive Function in Older Adults During Hospitalization: A Prospective Observational Study. *Health Sci Rep*. 2026;9(8):e72857. PMID 42504232
6. Jahan-Mihan A, Stevens P, Medero-Alfonso S, et al. The Role of Water-Soluble Vitamins and Vitamin D in Prevention and Treatment of Depression and Seasonal Affective Disorder in Adults. *Nutrients*. 2024;16(12):1902. PMID 38931257
7. Miller AH, Raison CL. The role of inflammation in depression: from evolutionary imperative to modern treatment target. *Nat Rev Immunol*. 2016;16(1):22-34. PMID 26711676
8. Abdelmessih GT, Bransby L, Cummins H, Jackson ML, Lim YY. Associations of self-reported obstructive sleep apnea with cognition and dementia risk in cognitively unimpaired middle-aged adults. *Alzheimers Dement*. 2026;22(7):e71553. PMID 42370433
9. Livingston G, Huntley J, Sommerlad A, et al. Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*. 2020;396(10248):413-446. PMID 32738937
10. Smith JR, Huang AR, Lin FR, Reed NS, Deal JA. The Population Attributable Fraction of Dementia From Audiometric Hearing Loss Among a Nationally Representative Sample of Community-Dwelling Older Adults. *J Gerontol A Biol Sci Med Sci*. 2023;78(7):1300-1306. PMID 37139951
11. Ruiz-Julian M, Piñol-Ripoll G. Anticholinergic burden and its relationship with cognitive impairment in a neuropsychiatric unit. *Farm Hosp*. 2026;50(4):T247-T252. PMID 42156261
12. Marc DT, Ailts JW, Campeau DC, Bull MJ, Olson KL. Neurotransmitters excreted in the urine as biomarkers of nervous system activity: validity and clinical applicability. *Neurosci Biobehav Rev*. 2011;35(3):635-644. PMID 20696183

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

If you are in crisis right now, call or text **988** (Suicide and Crisis Lifeline, 24/7). Food can wait; that call cannot.

Nothing here is a reason to stop, skip, or reduce a psychiatric medication. The strongest diet trial in this territory tested nutrition ADDED to standard care, never instead of it, and that is exactly how this guide is meant to be used.

This guide makes no dementia prevention or reversal claims. Where the research is observational or modest, it says so.

The one idea underneath everything

Key 8 of *Unlock Your Brain Health* says it plainly: the two interventions with the deepest evidence for brain fog and mood are the least exciting ones, what you eat on a normal Tuesday and whether you move regularly. Supplements layered on a poor foundation are the most common and expensive mistake in this territory. This guide is the foundation half, and it is deliberately boring in the way the evidence is boring.

Part one: the two trials worth knowing by name

MIND, for the aging brain. In a cohort of older adults followed for years, those whose eating most resembled the MIND pattern, a hybrid of Mediterranean and DASH emphasizing leafy greens, other vegetables, berries, nuts, whole grains, fish, and olive oil, declined cognitively at a substantially slower rate, a difference the researchers described as equivalent to being 7.5 years younger (PMID 26086182). Honest grade. Observational, an association rather than proof, and among the strongest and most consistent associations in this entire field.

SMILES, for mood. A randomized controlled trial tested a dietitian-supported Mediterranean-style dietary improvement program added to participants' existing depression care against a social-support control. The diet group improved significantly more, with roughly one in three reaching remission versus one in twelve in the control group (PMID 28137247). Honest grade. A real randomized trial, modest size, tested as an addition to standard care. That is the model: food as a treatment partner, not a replacement.

What the two trials agree on is almost the same plate: vegetables at most meals with leafy greens most days, berries regularly, fish and legumes as protein anchors, nuts and olive oil as default fats, whole grains over refined, and a sharp drop in ultra-processed food. Nobody's brain plate needs to be more exotic than that.

Part two: fuel stability, the every-three-hours story

Key 4's blood sugar biology translates to the plate simply. The brain runs poorly on fuel spikes and crashes, and the classic pattern, foggy and irritable late morning after a carbohydrate breakfast, better after regular balanced meals, is worth testing on yourself for two weeks:

1. **Protein at every meal, breakfast especially.** Eggs, fish, yogurt, legumes, meat: any of them beats toast alone.
2. **Never carbohydrate naked.** Pair it with protein, fat, or fiber to flatten the curve.
3. **Regular meal timing** while symptoms are active; skipped meals are a common, cheap-to-fix fog trigger.
4. **Front-load the day.** A substantial breakfast and lunch with a lighter dinner supports both fuel stability and sleep.

If the lab guide's glucose, HbA1c, or fasting insulin numbers were drifting, this part is doing double duty for brain and metabolism at once.

Part three: the specific foods with brain-specific reasons

Fish, especially the fatty ones. Salmon, sardines, trout, and herring supply EPA and DHA. The supplement guide covers the trial evidence for EPA in diagnosed depression; the food-first version is two fish meals a week, which is also the MIND pattern's floor.

Leafy greens most days and berries most weeks: the two food groups the MIND pattern weights hardest, and the easiest to systematize (a standing salad, frozen berries on breakfast).

Fermented and fiber-rich foods for the gut-brain terrain. Key 3's honest framing applies: the gut-brain axis is genuine science that is still maturing, and the defensible food translation is unspectacular. Fiber variety feeds the microbes that make short-chain fatty acids, and yogurt, kefir, and fermented vegetables are reasonable inclusions rather than miracle foods.

Coffee and tea, timed. Caffeine is fine for most brains and genuinely hostile to sleep when late; a noon cutoff is the cheap experiment for anyone with both fog and poor sleep.

Alcohol, honestly. It fragments sleep architecture even when it speeds falling asleep, and poor sleep is a direct fog generator (the ebook's Key 6 covers the glymphatic cleaning story). If evenings include alcohol and mornings include fog, that pairing deserves a two-week trial separation before any supplement does.

Hydration, plainly. Even mild dehydration reads as fatigue and poor concentration for many people; pale-straw urine is the practical gauge.

What this guide deliberately does not include

This guide leaves out “brain detox” diets, ketogenic-cures-Alzheimer's claims (the ebook's Key 4 explains why the “type 3 diabetes” nickname is shorthand for real research, not a diagnosis or a protocol), elimination diets without a specific tested reason, and any food list that promises to raise a neurotransmitter. The two trials in part one are the honest ceiling of what nutrition evidence currently supports here, and they are plenty.

What a day looks like

- **Breakfast.** Eggs with sauteed greens; oatmeal with frozen berries and walnuts on the side.
- **Lunch.** A big salad with salmon or lentils, olive oil, whatever vegetables exist.
- **Dinner.** Any protein, two vegetables, beans or a whole grain, lighter than lunch.
- **Rhythm.** Fish twice a week, greens most days, berries most weeks, caffeine before noon, alcohol watched honestly, water to pale straw.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and your medication list.

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Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

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If you would rather start on your own

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Read the companion documents alongside this one. **Understanding Your Brain Fog and Mood Labs** covers the treatable causes to rule out first, and **The Brain Health Supplement Guide** covers what genuinely helps beyond the plate.

References

All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Brain-Kit-Citation-Ledger.md`.

1. Morris MC, Tangney CC, Wang Y, et al. MIND diet slows cognitive decline with aging. *Alzheimers Dement*. 2015;11(9):1015-1022. PMID 26086182
2. Jacka FN, O'Neil A, Opie R, et al. A randomised controlled trial of dietary improvement for adults with major depression (the 'SMILES' trial). *BMC Med*. 2017;15(1):23. PMID 28137247

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

If you are in crisis right now, call or text **988** (Suicide and Crisis Lifeline, 24/7). No supplement has any role in a crisis.

Nothing in this guide is a reason to stop, skip, or reduce a psychiatric medication. Every entry below is written as a possible addition to standard care, coordinated with your prescriber, never as an alternative to it. Stopping psychiatric medication abruptly can cause genuine withdrawal effects and worse.

The interaction section below is the most important part of this guide. If you take an antidepressant of any kind, read it before anything else.

This guide makes no dementia prevention or reversal claims, and it names the trial that taught everyone why.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

The serotonin interactions that outrank everything else here

Two popular mood supplements have real pharmacology, which is exactly why they carry real risk alongside psychiatric medication.

St. John's Wort has documented, clinically significant interactions with SSRIs and other antidepressants, with oral contraceptives, with blood thinners, and with a long list of prescriptions, including the risk of serotonin syndrome when combined with serotonergic antidepressants (PMID 41901246). It also reduces the effectiveness of several medications by speeding their breakdown, which is a quiet way to lose contraception or anticoagulation without noticing.

5-HTP raises serotonin directly, and combining it with an SSRI, SNRI, or MAOI carries a real risk of serotonin syndrome, with agitation, rapid heart rate, tremor, and in severe cases a medical emergency (PMID 30599219). **SAMe** sits in the same caution category alongside antidepressants.

The rule, with no exceptions this guide will endorse. None of these three starts, continues, or combines with a psychiatric prescription without your prescriber's explicit sign-off, and even without a prescription in the picture, they belong in an ask-first conversation, not a cart. If you develop agitation, racing heart, tremor, sweating, or confusion on any serotonin-active combination, that is an emergency, not a side effect to push through.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent fish oil or a B vitamin, so the evidence hierarchy over-documents patentable drugs and under-documents nearly everything else. A Cochrane methodology review found manufacturer sponsorship produces more favourable results and conclusions than independent funding, through a bias standard quality checks do not catch (PMID 28207928). So when this guide says “no evidence,” it will say whether it means *tested and failed*, *never properly tested*, or *tested small and won*. The asymmetry earns nutrients humility, not automatic credit: small nutrient trials carry their own publication bias toward positive findings. One rule, both directions. This category also has a unique stake in the rule, because the brain supplement aisle is the industry’s most profitable fog.

Tier 1: what actually has human evidence

EPA-predominant omega-3, for diagnosed depression, alongside care

The omega-3 literature looks contradictory until one distinction is applied, and then it organizes. A meta-analysis of 35 randomized trials found that in people with diagnosed depression, formulations that were predominantly EPA (more than half EPA) showed clinically meaningful benefit over placebo, while DHA-predominant formulations did not, and omega-3 did not prevent depressive symptoms in people without depression (PMID 27103682).

Three honest boundaries follow directly from that finding. This is evidence for **EPA-predominant products** specifically, so read the label ratio rather than the word “fish oil.” It is evidence in **diagnosed depression as an addition to care**, coordinated with whoever manages that care. And it is **not** evidence for prevention or for fog without mood symptoms, where the food-first version (two fish meals weekly, per the diet guide) remains the defensible play. Practical note: omega-3 at supplement doses has a blood-thinning effect worth naming to your prescriber if you take anticoagulants or face surgery.

Correcting measured deficiencies in B12, folate, and vitamin D

This tier's least glamorous entry is its most defensible: when the lab guide's workup finds a genuine deficiency, correcting it is evidence-based medicine, not supplementation culture. B12 deficiency produces neuropsychiatric symptoms that respond to repletion (PMID 35337631); folate status relates to depression risk in a U-shape that argues for sufficiency, not megadosing (PMID 41824854); and vitamin D is worth checking and correcting if low, with the review literature candid that supplementation trials for mood in non-deficient people are inconsistent (PMID 38931257). Forms matter modestly (methylated B vitamins for those who need them; vitamin D with food), doses follow the actual lab result, and retesting closes the loop. What this tier is not: a reason to take any of these at high doses with normal labs.

Tier 2: tested and failed, named honestly

Ginkgo biloba, for the aging brain

Ginkgo is the best-selling memory herb in the world, and it received exactly the trial its popularity deserved. The GEM trial randomized 3,069 older adults against placebo and followed them for a median of six years. Ginkgo did not reduce the incidence of dementia or Alzheimer's disease (PMID 19017911). That is what *tested and failed* looks like at full scale, on public money (the trial was funded by the NIH). If ginkgo is in your cabinet for memory protection, that is the study to weigh it against.

The nootropic blend shelf

Multi-ingredient “brain support,” “focus,” and “nootropic” stacks are the supplement industry’s fastest-growing fog. Their individual ingredients range from *never properly tested* to *tested small*, their combinations are essentially never tested at all, and blend labels routinely underdose the one ingredient with any evidence behind it. The same rule applies as in the rest of this product line. Every ingredient is chosen on purpose, at a dose you know, for a reason you can say out loud. A blend fails that rule by construction.

What actually protects the brain is not in a capsule

The honest close of this guide points back out of it. The strongest interventions in this territory are the diet pattern and the movement habit (the diet guide carries their trials), sleep treated as the non-negotiable it is, hearing checked and corrected, blood pressure and blood sugar managed, and the treatable causes in the lab guide actually ruled out. The Lancet Commission’s modifiable dementia risk factors are lifestyle and medical factors, not supplements (PMID 32738937). A reader who does those things and takes nothing from this guide’s shelves is better protected than one who inverts the order.

What I deliberately left out

Racetams and other gray-market “smart drugs,” high-dose single amino acids for neurotransmitter loading, “brain detox” products, and every blend whose mechanism is described in marketing language rather than chemistry. A guide reaches you without a history, a medication list, an exam, or any way for me to change course next week, so the bar for what goes in it is higher than the bar for a conversation in my office. None of those clear it.

Where to find quality versions of these

Quality varies; look for independent third-party testing (NSF, USP, or equivalent) whichever source you use. I maintain a professional dispensary at Root Health through Fullscript:

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. Nothing here was included, excluded, ranked, or dosed based on what it earns. Notice that the most powerful tools in this guide, sleep, movement, the diet pattern, a hearing check, and the 988 line when it is needed, cost nothing and earn me nothing. That is where I want you to start anyway.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and your full medication list, which the interaction section above should make feel non-optional.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own:

Free tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your Brain Fog and Mood Labs** covers the treatable causes that outrank every supplement here, and **The Brain Health Diet Guide** covers the two levers with the strongest evidence in this entire territory.

References

All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in [Brain-Kit-Citation-Ledger.md](#).

1. Caus MN, Lupoae M, Chitescu CL. Efficacy and Safety of Herbal Supplements with Anxiolytic, Antidepressant, and Sedative Action: A Review of Clinical Data and Toxicological Risks. *Pharmaceuticals (Basel)*. 2026;19(3):399. PMID 41901246
2. Dome P, Tombor L, Lazary J, Gonda X, Rihmer Z. Natural health products, dietary minerals and over-the-counter medications as add-on therapies to antidepressants in the treatment of major depressive disorder: a review. *Brain Res Bull*. 2019;146:51-78. PMID 30599219
3. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev*. 2017;2(2):MR000033. PMID 28207928
4. Hallahan B, Ryan T, Hibbeln JR, et al. Efficacy of omega-3 highly unsaturated fatty acids in the treatment of depression. *Br J Psychiatry*. 2016;209(3):192-201. PMID 27103682
5. Sahu P, Thippeswamy H, Chaturvedi SK. Neuropsychiatric manifestations in vitamin B12 deficiency. *Vitam Horm*. 2022;119:457-470. PMID 35337631
6. Zeng W, Ma H, Wei S, et al. Red blood cell folate and depression risk in U.S. women: A dose-response analysis from a nationwide cross-sectional study (2009-2018). *Medicine (Baltimore)*. 2026;105(11):e48023. PMID 41824854
7. Jahan-Mihan A, Stevens P, Medero-Alfonso S, et al. The Role of Water-Soluble Vitamins and Vitamin D in Prevention and Treatment of Depression and Seasonal Affective Disorder in Adults. *Nutrients*. 2024;16(12):1902. PMID 38931257
8. DeKosky ST, Williamson JD, Fitzpatrick AL, et al. Ginkgo biloba for prevention of dementia: a randomized controlled trial. *JAMA*. 2008;300(19):2253-2262. PMID 19017911
9. Livingston G, Huntley J, Sommerlad A, et al. Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*. 2020;396(10248):413-446. PMID 32738937

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Educational content only. Not medical advice. Not a diagnosis. Never stop, skip, or change a psychiatric medication based on this guide. Serotonin syndrome symptoms on any supplement-medication combination are an emergency. In crisis, call or text 988 now.